

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 3

**Duration : 1 Hour
Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I K.AISHWARYA(192572116) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

Step 1: Define Variables and Domains

The variables are:

- **D1** → Morning, Afternoon, Night
- **D2** → Morning, Afternoon, Night
- **D3** → Morning, Afternoon, Night

Considering the individual restrictions:

- **D1** → **Morning, Afternoon**
- **D2** → **Morning, Afternoon, Night**
- **D3** → **Afternoon, Night**

Step 2: Start Backtracking

We assign shifts one doctor at a time.

Try D1 = Morning

Check:

- D1 cannot work Night → **Satisfied ✓**
- No other doctor is assigned Morning yet → **Satisfied ✓**

Current assignment:

D1 = Morning

Step 3: Assign D2

Try:

D2 = Morning

Constraint check:

- Morning is already occupied by D1.
- Only one doctor per shift.

Therefore:

D2 = Morning → X Invalid

Backtrack

Try the next possible shift for D2:

D2 = Afternoon

Check:

- Afternoon is free → ✓
- D2 must work before D3 → D3 must be after Afternoon → ✓

Current assignment:

D1 = Morning

D2 = Afternoon

Step 4: Assign D3

D3 cannot work Morning.

Available shifts:

- Morning → Already occupied by D1
- Afternoon → Already occupied by D2
- Night → Available

Therefore:

D3 = Night

Check:

- D3 cannot work Morning → ✓
- D2 must work before D3 → Afternoon < Night → ✓
- Night is free → ✓

Current assignment:

D1 = Morning

D2 = Afternoon

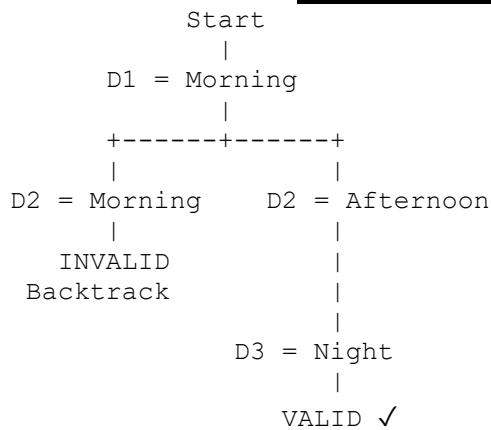
D3 = Night

Step 5: Final Constraint Checking

Constraint	Check	Result
D1 cannot work Night	D1 = Morning	✓
D2 must work before D3	Afternoon < Night	✓
Only one doctor per shift	No repeated shifts	✓
D3 cannot work Morning	D3 = Night	✓
All doctors assigned exactly one shift	D1, D2, D3 assigned	✓

All constraints are satisfied.

Backtracking Search Tree



Final Valid Schedule

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

Final Answer

D1 → Morning
D2 → Afternoon
D3 → Night

Thus, using **Backtracking Search**, the valid assignment is **Morning → D1, Afternoon → D2, Night → D3**.

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- Movement allowed: Up, Down, Left, Right
- Cost of each move = 1
- Cannot pass through obstacles
- Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion

and priority queue updates and determine the optimal path and cost.

Given 5×5 Grid

Let:

- **S** = Start
- **G** = Goal
- **#** = Obstacle
- **.** = Free cell

	1	2	3	4	5
1	S . . # .				
2	. # . # .				
3	. # . . .				
4	. . # # .				
5	 G				

Movement: Up, Down, Left, Right

Cost of each move: 1

Heuristic: Manhattan Distance

$$h(n)=|x_n-x_G|+|y_n-y_G| \quad h(n)=|x_n-x_G|+|y_n-y_G|$$

For the goal at **(5,5)**:

Step 1: Start Node

Start:

$$S = (1,1)$$

Manhattan distance:

$$h(S)=|1-5|+|1-5|=8 \quad h(S)=|1-5|+|1-5|=8$$

Priority queue:

[(1,1)]

Expand **(1,1)**.

Possible movements:

- Up → outside grid X
- Left → outside grid X
- Right → (1,2) ✓
- Down → (2,1) ✓

Queue becomes:

[(1,2) , (2,1)]

Step 2: Expand (1,2)

For (1,2):

$$h = |1-5| + |2-5| = 7 \quad h = |1-5| + |2-5| = 7$$

Expand (1,2).

Possible nodes:

- Right $\rightarrow (1,3)$ ✓
- Down $\rightarrow (2,2)$ = obstacle ✗
- Left $\rightarrow (1,1)$ already visited ✗
- Up $\rightarrow$ outside ✗

Priority queue:

[(2,1) , (1,3)]

Step 3: Expand (2,1)

For (2,1):

$$h = |2-5| + |1-5| = 7 \quad h = |2-5| + |1-5| = 7$$

Possible movements:

- Down $\rightarrow (3,1)$ ✓
- Up $\rightarrow (1,1)$ visited ✗
- Right $\rightarrow (2,2)$ obstacle ✗
- Left $\rightarrow$ outside ✗

Priority queue:

[(1,3) , (3,1)]

Step 4: Expand (1,3)

For (1,3):

$$h = |1-5| + |3-5| = 6 \quad h = |1-5| + |3-5| = 6$$

Possible movements:

- Down $\rightarrow (2,3)$ ✓
- Right $\rightarrow (1,4)$ obstacle ✗
- Left $\rightarrow$ visited
- Up $\rightarrow$ outside

Queue:

[(3,1) , (2,3)]

Step 5: Expand (3,1)

For (3,1):

$$h=|3-5|+|1-5|=6 \quad h=|3-5|+|1-5|=6$$

Possible:

- Down $\rightarrow$ (4,1) ✓
- Up $\rightarrow$ visited
- Right $\rightarrow$ (3,2) obstacle X
- Left $\rightarrow$ outside

Queue:

[(2,3) , (4,1)]

Step 6: Expand (2,3)

$$h=|2-5|+|3-5|=5 \quad h=|2-5|+|3-5|=5$$

Possible:

- Down $\rightarrow$ (3,3) ✓
- Up $\rightarrow$ (1,3) visited
- Right $\rightarrow$ (2,4) obstacle X
- Left $\rightarrow$ (2,2) obstacle X

Queue:

[(4,1) , (3,3)]

Step 7: Continue Expansion

The search continues through the free cells:

(3,3)
 $\downarrow$
(3,4)
 $\downarrow$
(3,5)
 $\downarrow$
(4,5)
 $\downarrow$
(5,5) = G

The corresponding Manhattan distances decrease toward the goal.

Priority Queue Summary

Step	Expanded Node	New Nodes Added	Priority Queue
1	(1,1)	(1,2), (2,1)	(1,2), (2,1)

Step	Expanded Node	New Nodes Added	Priority Queue
2	(1,2)	(1,3)	(2,1), (1,3)
3	(2,1)	(3,1)	(1,3), (3,1)
4	(1,3)	(2,3)	(3,1), (2,3)
5	(3,1)	(4,1)	(2,3), (4,1)
6	(2,3)	(3,3)	(4,1), (3,3)
7	(3,3)	(3,4)	...
8	(3,4)	(3,5)	...
9	(3,5)	(4,5)	...
10	(4,5)	(5,5)	G

Optimal Path

The optimal path is:

S
 ↓
 (2, 1)
 ↓
 (3, 1)
 ↓
 (4, 1)
 ↓
 (4, 2)
 ↓
 (5, 2)
 ↓
 (5, 3)
 ↓
 (5, 4)
 ↓
 G

Path Representation

S → ↓ → ↓ → ↓ → → ↓ → → → G

Total Cost

There are **8 moves**.

Since each move costs **1**:

Total Cost=8×1=8

Final Answer

Optimal path:

S → (2,1) → (3,1) → (4,1) → (4,2) → (5,2) → (5,3) → (5,4) → G

Optimal cost = 8

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and

rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

The rescue robot works in a **partially observable grid environment**. Each constraint affects its decisions as follows:

1. **Movement** **constraints**
The robot can move only **Up, Down, Left, or Right**. It cannot move diagonally, so it must select one of these four directions at every step.
2. **Obstacles**
Blocked cells cannot be entered. The robot must identify obstacles using its sensors and find another safe route.
3. **Limited** **sensing**
The robot can observe only **adjacent cells**. It does not know the complete map in advance. Therefore, it must make decisions using its current knowledge and update the map as it moves.
4. **Movement** **cost**
Every normal movement has a cost of **1**. The robot should avoid unnecessary movements because they increase the total cost.
5. **Risky** **zones**
Entering a risky zone adds an extra cost of **2**. Therefore, entering such a cell has a total cost of:

$$1 + 2 = 3$$

The robot should prefer a longer safe route if it has a significantly lower cost than a risky route.

6. **Cost** **minimization**
The robot must choose a path with the **lowest total cost**, rather than simply choosing the shortest path.
7. **Dynamic** **knowledge**
As the robot moves, it discovers new obstacles and risky areas. It must update its knowledge and **re-plan its path** when new information is obtained.

Conclusion

These constraints require the robot to continuously **sense, decide, move, update its knowledge, and re-plan**. Therefore, the robot needs an intelligent search strategy that considers both **path cost and newly discovered information**.

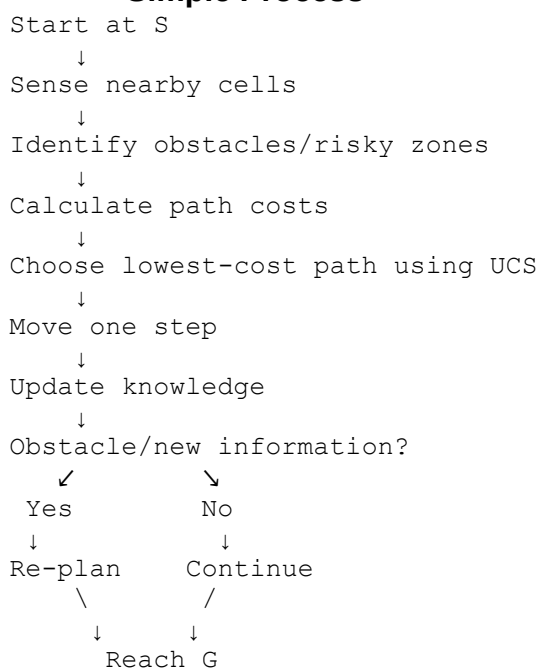
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

For this problem, **Uniform Cost Search (UCS) combined with an Online Search Agent** is suitable.

Steps involved:

1. **Start at S**
The robot begins at the starting position S.
2. **Sense the surroundings**
The robot observes the adjacent cells to identify:
 - Free paths
 - Obstacles
 - Risky zones
3. **Calculate movement cost**
 - Normal cell $\rightarrow$ cost = 1
 - Risky cell $\rightarrow$ cost = 1 + 2 = 3
 - Obstacle $\rightarrow$ cannot enter
4. **Use Uniform Cost Search**
UCS selects the available path with the **lowest accumulated cost**.
5. **Move to the selected cell**
The robot moves along the lowest-cost path.
6. **Update knowledge**
After every movement, the robot observes new adjacent cells and updates its internal map.
7. **Re-plan when necessary**
If a newly discovered obstacle blocks the planned path, the robot searches for another lowest-cost path.
8. **Continue until G is reached**
The process continues until the robot reaches the survivor at G.

Simple Process



Result

The robot dynamically finds a **minimum-cost and safer route** to the survivor while adapting to information discovered during navigation.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

1. Intelligent Agent

The rescue robot is an **intelligent agent** because it:

- **Perceives** the environment using sensors.
- **Processes** the information it receives.
- **Makes decisions** about which direction to move.
- **Acts** by moving through the building.
- **Updates its knowledge** when it discovers new obstacles or risky areas.
- **Works toward a goal**, which is reaching the survivor at **G**.

Example:

If the robot detects an obstacle in front of it, it senses the obstacle and chooses another available direction.

2. Rationality

The robot is a **rational agent** because it chooses actions that help achieve the goal with the **lowest possible cost while maintaining safety**.

For example:

- Normal movement → cost 1
- Risky-zone movement → cost 3
- Obstacle → cannot be entered

If two paths are available, the robot chooses the path with the **lower total cost and safer conditions**.

Thus, rationality means choosing the **best action based on the information currently available**.

3. Environment Type

The rescue building environment has several AI environment characteristics:

Environment Property	Description
Partially observable	Robot can see only nearby cells
Sequential	Current movement affects future decisions
Dynamic	New obstacles/risk information can be discovered
Discrete	The building is represented as separate grid cells
Uncertain	Robot does not initially know the complete environment
Goal-based	Main goal is to reach the survivor safely

Conclusion

The robot acts as an **intelligent and rational agent** operating in a **partially observable, sequential, dynamic, discrete, and uncertain environment**.

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

The combination of **Uniform Cost Search (UCS)** and **Online Search** is suitable for this rescue robot because the environment is **partially observable** and movement costs are not always equal.

Reasons:

1. **Handles different costs**
UCS considers the actual cost of each path. Normal movement costs **1**, while entering a risky zone costs **3**.
2. **Finds the minimum-cost path**
UCS expands the path with the lowest total cost first, helping the robot find a cost-efficient route.
3. **Works with obstacles**
Blocked cells are excluded from the search, so the robot can find an alternative route.
4. **Suitable for limited information**
The robot does not know the complete environment initially. Online search allows it to make decisions using the information currently available.
5. **Updates dynamically**
When the robot discovers a new obstacle or risky zone, it can update its map and search for a new path.
6. **Supports safe navigation**
The robot can consider risky-zone costs and avoid expensive or unsafe routes when a better alternative exists.

Final Justification

Therefore, **UCS + Online Search Agent** is the most suitable approach because it allows the robot to **adapt to an unknown environment, avoid obstacles, consider different movement costs, and reach the survivor using a minimum-cost safe path.**

e) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

Since this is the repeated version of **(b)**, the same **Uniform Cost Search (UCS) with Online Search Agent** approach can be used.

Steps:

1. **Initialize the robot**
 - Start from position **S**.
 - Set the initial path cost to **0**.
2. **Observe nearby cells**
 - Sense the adjacent cells.
 - Identify free cells, obstacles, and risky zones.
3. **Assign costs**
 - Normal movement = **1**
 - Risky zone = **3** (1 movement cost + 2 additional risk cost)
 - Obstacle = **Not allowed**
4. **Apply UCS**
 - Place possible paths in a priority queue.

- Select the path having the **lowest total cost**.
- 5. **Move**
 - Move to the selected cell.
 - Add its movement cost to the total path cost.
- 6. **Update the map**
 - Sense the surroundings again.
 - Add newly discovered obstacles and risky zones to the robot's knowledge.
- 7. **Re-plan**
 - If the current path becomes blocked or unsafe, run UCS again using the updated information.
- 8. **Reach the goal**
 - Continue sensing, planning, moving, and updating until the robot reaches **G**.

Overall Process

S → Sense → Calculate Cost → UCS → Move → Update Knowledge → Re-plan if needed → G

Thus, the robot continuously adapts its path according to the **latest information and minimum accumulated cost**.

f) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
 - Rationality
 - Environment type
- 1. Intelligent Agent**

The rescue robot acts as an **intelligent agent** because it:

- Senses its surroundings.
- Collects information about obstacles and risky zones.
- Makes decisions about which direction to move.
- Takes action by moving through the grid.
- Updates its knowledge after every movement.
- Works toward the goal of reaching **G**.

Example: If the robot detects an obstacle ahead, it senses it and chooses another available path.

2. Rationality

The robot behaves **rationally** by selecting the action that best achieves its goal with minimum cost and safe navigation.

- Normal movement = **1**
- Risky-zone movement = **3**
- Obstacle = **Cannot enter**

The robot compares available paths and chooses the one with the **lowest total cost** based on the information it currently has.

3. Environment Type

The robot operates in the following environment:

Environment Type	Explanation
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Environment Type	Explanation
Partially Observable	It can observe only nearby cells
Sequential	Each movement affects future decisions
Dynamic	New information can be discovered while moving
Discrete	The environment consists of separate grid cells
Uncertain	The complete environment is initially unknown
Goal-Based	The goal is to reach the survivor at G

Conclusion

The robot is an **intelligent and rational agent** that operates in a **partially observable, sequential, dynamic, discrete, and uncertain environment** and continuously adapts its actions to reach the survivor safely.

g) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

The combination of **Uniform Cost Search (UCS)** and **Online Search** is suitable for the rescue robot because the robot has **limited information and different movement costs**.

Reasons:

1. **Handles different costs**
UCS considers the cost of every movement. A normal cell costs **1**, while a risky cell costs **3**.
2. **Finds the minimum-cost path**
UCS selects the path with the lowest accumulated cost, helping the robot reach the survivor efficiently.
3. **Handles obstacles**
Obstacles cannot be entered, so the robot searches for alternative paths.
4. **Works with partial information**
Online search allows the robot to make decisions using the information it currently knows.
5. **Updates dynamically**
When a new obstacle or risky zone is detected, the robot updates its knowledge and can **re-plan its route**.
6. **Supports safe navigation**
The additional cost for risky zones encourages the robot to prefer safer paths when they are more cost-effective.

Final Justification

Therefore, **UCS with an Online Search Agent** is the most suitable approach because it allows the rescue robot to **adapt to an unknown environment, avoid obstacles, consider different costs, and reach the survivor through a minimum-cost and safe route**.

